

EXPERIENCE

Senior Theoretical Researcher — Institute for Advanced Study, Princeton1933 – 1955

- Built mathematical models of gravitation and field theory, translating ambiguous physical observations into testable quantitative frameworks.
- Led independent research programs across relativity, quantum theory, and statistical physics while collaborating with international scientific teams.
- Communicated complex technical ideas through papers, lectures, and public-facing writing for expert and non-expert audiences.

Skills: Mathematical Modeling, Research Strategy, Scientific Communication, Cross-Functional Collaboration

Professor of Physics — University of Berlin, Berlin1914 – 1933

- Developed the general theory of relativity, creating a predictive model that explained gravitational lensing, orbital precession, and cosmological dynamics.
- Designed thought experiments and analytical workflows to challenge assumptions, reduce hidden variables, and surface falsifiable predictions.
- Mentored researchers and coordinated seminar discussions across theoretical physics, astronomy, and applied mathematics.

Skills: Theory Design, Problem Decomposition, Mentorship, Systems Thinking

Technical Expert — Swiss Patent Office, Bern1902 – 1909

- Reviewed electromechanical patents for novelty, implementation feasibility, and technical clarity under strict documentation standards.
- Evaluated synchronization, signal transmission, and measurement systems, improving ability to reason from engineering constraints to core principles.
- Produced concise technical decisions for high-volume applications while maintaining accuracy and repeatable review criteria.

Skills: Technical Review, Electromechanical Systems, Documentation, Prior-Art Analysis

SELECTED PROJECTS

- General Relativity** – Unified acceleration and gravity into a geometric theory with measurable astronomical predictions.
- Photoelectric Effect** – Modeled light quanta to explain electron emission behavior and support modern quantum theory.
- Brownian Motion** – Converted microscopic particle motion into statistical evidence for atomic theory.
- Mass-Energy Equivalence** – Derived a compact relationship connecting mass and energy across physical systems.

TECHNICAL SKILLS

Core: Mathematical modeling, analytical reasoning, uncertainty reduction, hypothesis testing

Domains: Relativity, quantum theory, statistical mechanics, electrodynamics, cosmology

Methods: Differential geometry, tensor calculus, stochastic processes, dimensional analysis

Tools: Technical writing, patent review, lecture design, peer review, public communication

Languages: German, English, French, Italian

AWARDS & HONORS

- Nobel Prize in Physics** – Awarded for services to theoretical physics and work on the photoelectric effect (1921)
- Copley Medal** – Royal Society recognition for outstanding scientific achievement (1925)

EDUCATION

University of Zurich1905

Ph.D. in PhysicsZurich, Switzerland

Swiss Federal Polytechnic1896 – 1900

Diploma in Mathematics and PhysicsZurich, Switzerland

SELECTED PUBLICATIONS

- Einstein, A. (1915). The Field Equations of Gravitation.
- Einstein, A. (1905). On a Heuristic Point of View Concerning the Production and Transformation of Light.
- Einstein, A. (1905). On the Electrodynamics of Moving Bodies.